MILITARY SPECIFICATION

MOUNTING-PLENUM MT-2243(*)/ASQ-56

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements for the Mounting-Plenum MT-2243/ASQ-56. When installed as part of Integrated Electronic Central AN/ASQ-56 this equipment shall provide the following:

(1) Equipment mounting facilities

(2) Vibration isolation

(3) Facilities for electrical connections

(4) Distribution of cooling air to equipments

1.2 Associated Equipments - The equipment shall be capable of operating with the following associated equipment which is not to be supplied as part of this equipment.

Item	Type Designation
Receiver-Transmitter, Radio	RT-541/ASQ
Decoder, Pulse	KY-309/ASQ
Coder-Receiver-Transmitter	KY-308/ASQ
Amplifier-Power Supply-Receiver	AM-2310/ASQ
Receiver-Transmitter, Radio	RT-542/ASQ

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases, Bases; Mounting; and Mounts, Vibration (for use with Electronic Equipment in Aircraft)
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment

FSC 5340

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MIL-I-22668 Integrated Electronic Centrals AN/ASQ-56, AN/ASQ-57 and AN/ASQ-58

2.2 Availability of Documents -

(1) When requesting specifications, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing parts and material requirements shall conform to Specification MIL-I-22668.

3.2 Design and Construction

3.2.1 General - This equipment shall conform with the applicable requirements of Specification MIL-I-22668 for design, construction, and workmanship except in the paragraphs as modified below:

3.2.5.1 Cables and Connectors - No cables or connectors shall be required as part of the equipment, however, provision shall be made in the rear of the equipment for mounting applicable mating connectors of equipments listed below:

Type Designation	Mating Connector
RT-541/ASQ	Amphenol 126-20572
KY-309/ASQ	Amphenol 126-20573
KY-308/ASQ	Amphenol 126-20572
AM-2310/ASQ	Amphenol 126-20573
RT-542/ASQ	Amphenol 126-20572 (or approved equivalents)

3.2.12 Maximum Altitude - The maximum altitude under service conditions for the equipment shall be 75,000 feet.

3.2.12.1 Cooling Air Flow Requirements - Air ducting shall accomodate maximum flow quantities varying from 1.5 to 5.9 pounds per minute. Air ducting shall accommodate cooling air pressures with respect to the ambient varying between 1.26 and 2.28 psi. Leakage from the ducts, plenum, and seals shall not exceed 10% of the total quantity at the flow rate of 1.5 pounds per minute (1.26 psi pressure). The air inlet pipe shall be of one-piece construction (except for flange) such that a continuous bend is obtained.

3.2.12.2 Vibration - The equipment, and those listed in 1.2 must operate satisfactorily in accordance with the vibration range input curve of figure 1, with resonance and cycling tests in accordance with Specification MIL-T-5422. The transmitted acceleration

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from the mounting-plenum to the mounted equipment shall not exceed 1-1/2 G's. No resonant or random peak accelerations shall exceed 2 G's over the specified frequency range of figure 1. The vibration isolators shall show no mechanical damage after one hour excitation at the system natural frequency in each of three mutually perpendicular planes of motion, with the specified input amplitudes and/or acceleration as given in range curve of figure 1.

3.3 Detail Requirements:

3.3.1 Mounting-Plenum MT-2243/ASQ-56 - The mounting-plenum shall meet the following additional requirements:

3.3.1.1 Form Factor - The mounting-plenum shall be contained in a single unit capable of being directly mounted to the aircraft structure using ordinary hand tools. The overall dimensions shall not exceed 6-13/16 by 46-1/4 by 24-3/8 inches.

3.3.1.2 Weight - The weight of the mounting-plenum shall not exceed 35 pounds.

3.3.1.3 Extractor Bushings - Equipment extractor bushings shall be threaded 1/4-16 NA-2G and fabricated from beryllium copper heat treated to 20 Rockwell C minimum.

3.3.1.4 Guide Pins - Guide pins at the front of the mounting-plenum shall be of stainless steel and shall not interfere with the seating of the extractor into the extractor bushing.

3.3.1.5 Inserts - The guide pin holes at the rear of the mounting-plenum shall have stainless steel inserts.

3.3.1.6 Access Holes - Access holes shall be provided to facilitate extractor bushing replacement in the front channel of the mounting-plenum. Access holes shall be provided if necessary to provide replacement of ground nuts.

3.3.1.7 Ground Nuts - Six 10-32 ground nuts shall be provided at the rear of the mounting-plenum for equipment grounding.

3.3.1.8 Captive Stop Nuts - No captive stop nuts of the elastic or clinch types shall be used.

3.3.1.9 Equipment Connector Engaging Forces - The rear connector mounting brackets shall be capable of withstanding a connector engaging force of 75 pounds each without distortion.

3.3.1.10 Seals - A ring seal shall be fastened to or made a part of a metal surface which is below the top surface of the slide rails supporting the equipment. The ring seal shall provide the connection from the equipment unit to the plenum in the mount. The seal shall be free to deflect into the air opening in the plenum such that when sealing off the equipment it does not prevent the equipment from seating on the guide rails of the mount. The seal shall be made of neoprene and cloth covered on the sealing surface to reduce drag forces and wear abrasion of the seal. Seal must be removable for replacement.

3.3.1.11 Screw Assemblies - Assembly screws must be tight. Number 6 hardware shall be tightened to a torque of 10 inch pounds, number 8 hardware, 20 inch pounds.

3.3.1.12 Silk Screening - Nomenclature of associated equipment listed in paragraph 1.2 shall be silk screened on the bottom of the mounting-plenum as far forward as possible, and such that the Lettering is upright when the bottom of the mounting-plenum is viewed with the forward channel at the top.

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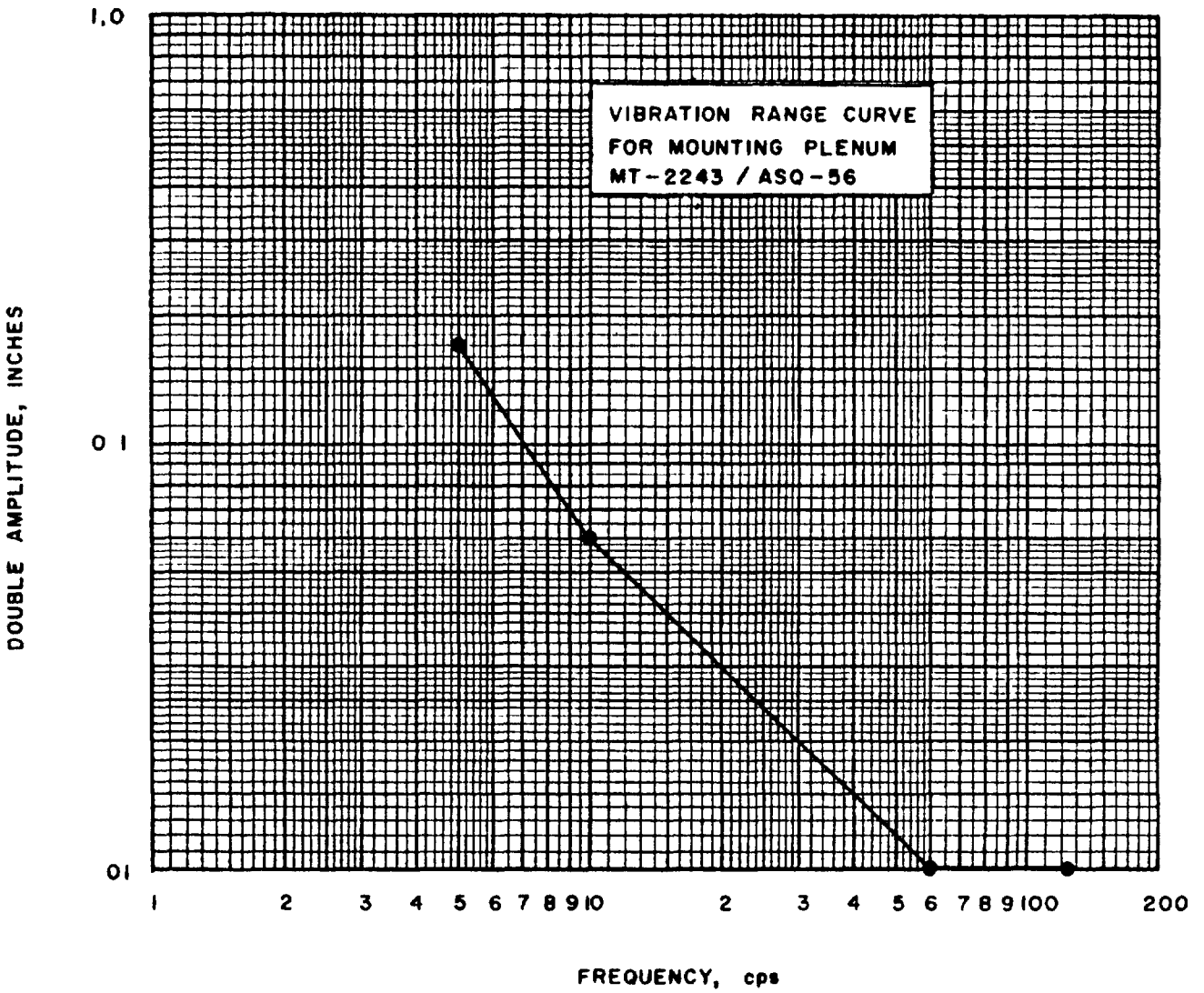


Figure 1

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3.3.1.13 Electrical Bonding - Electrical bonding shall be in accordance with Specification MIL-C-172.

3.3.1.14 Vibration Isolators - Mounting-Plenum MT-2243/ASQ-56 consists of a plenum and the following four isolators:

Barry Controls Part Number 6375-2	Right Front Assembly
Barry Controls Part Number 6375-1	Left Front Assembly
Barry Controls Part Number 6376	Right Rear Assembly
Barry Controls Part Number 6777 (or approved equivalents)	Left Rear Assembly

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the test requirements of Specification MIL-I-22668 to determine compliance with all the applicable requirements.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-I-22668.

6. NOTES

6.1 Precedence - When the requirements of the contract, this specification, applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) System Specification - Integrated Electronic Centrals AN/ASQ-56, AN/ASQ-57, AN/ASQ-58 Specification MIL-I-22668 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specification where applicable, shall be approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.2 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 of MIL-I-22668. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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MIL-M-22673(AS)
AMENDMENT-2
15 FEBRUARY 1968
Superseding
AMENDMENT-1
15 OCTOBER 1967

MILITARY SPECIFICATION

MOUNTING PLENUMS MT-2243(*)/ASQ-56 AND MT-3946(*)/ASQ-56B

This amendment forms a part of Military Specification MIL-M-22673(WEP) dated 1 November 1960 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-M-22673(WEP) is changed to MIL-M-22673(AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

Page 1, Title: Delete and substitute the title of this amendment.

Page 1, Paragraph 1.1: Delete the first two sentences and substitute "This specification covers the requirements for Mounting Plenums MT-2243/ASQ-56 and MT-3946/ASQ-56B. These mounting plenums, when installed as part of their respective centrals, shall provide the following:"

Page 1, paragraph 1.2: Delete and substitute: "1.2 Associated Equipment The equipment shall be capable of operating with the following associated equipment which is not to be supplied as part of the equipment covered by this specification.

ITEM	TYPE DESIGNATION
Receiver-Transmitter	RT-384/ARN-52 mounted on MX-4773/ASQ-56
Coder-Receiver-Transmitter	KY-308/ASQ See Note 1
Coder-Receiver-Transmitter	KY-533A/ASQ See Note 2
Amplifier-Power Supply-Receiver	AM-2310/ASQ
Receiver-Transmitter,Radio	RT-542 See Note 1
Receiver-Transmitter,Radio	RT-542A/ See Note 2"

Page 1, Bottom of Page: Add

"Note 1: This equipment associated with the MT-2243/ASQ-56 only

Note 2: This equipment associated with the MT-3946/ASQ-56B only."

Page 2, paragraph 2.1: Delete

"MIL-I-22668	Integrated Electronic Centrals AN/ASQ-56, AN/ASQ-57, and AN/ASQ-58."

Add:

"MIL-I-81007	Integrated Electronic Centrals AN/ASQ-56A and AN/ASQ-56B."

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Page 2, paragraph 3.1: Delete

"MIL-I-22668" and substitute "MIL-I-81007"
Make this substitution wherever MIL-I-22668 appears in this specification.

Page 2, paragraph 3.2.5.1: Delete all after "listed below" and substitute

"Type Designation	Mating Connector
MX-4773/ASQ-56	Amphenol 126-20572
	Amphenol 126-20573
KY-308/ASQ	Amphenol 126-20572
AM-2310/ASQ	Amphenol 126-20573
RT-542/ASQ or RT-542A/ASQ	Amphenol 126-20572
KY-533A	Amphenol 126-20572
	Cannon DDD-505(053669-0025)
	(or approved equivalents)"

Page 2, paragraph 3.2.12.1: Delete

"1.5 to 5.9 pounds per minute." and substitute "90 to 354 pounds per hour."

Page 3, paragraph 3.3.1: Delete and substitute

"3.3.1 Mounting Plenums MT-2243/ASQ-56 and MT-3946/ASQ-56B The mounting plenums shall meet the following additional requirements:"

Page 3, paragraph 3.3.1.1: Delete all after "exceed' and substitute

"$6\frac{7}{8}$ by $46\frac{3}{8}$ by $24\frac{1}{2}$ inches for the MT-2243/ASQ-56 and $7\frac{5}{8}$ by $46\frac{3}{8}$ by $24\frac{1}{2}$ inches for the MT-3946/ASQ-56B."

Page 3, paragraph 3.3.1.9: Delete "75" and substitute "100".

Page 4, graph title: Add "and MT-3946/ASQ-56B".

Page 5, paragraph 3.3.1.14: Delete "Mounting-Plenums MT-2243/ASQ-56 consists" and substitute "Mounting-Plenums MT-2243/ASQ-56 and MT-3946/ASQ-56B consist".

Page 5, paragraph 6.1 (2): Delete and substitute

"(2) System Specification - Integrated Electronic Central AN/ASQ-56A and AN/ASQ-56B, Specification MIL-I-81007 shall have procedence over this specification."

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